TP 1504

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Soit le modèle relationnel suivant relatif à la gestion des notes annuelles d'une promotion d'étudiants :

- ETUDIANT(NEtudiant, Nom, Prénom)
- MATIERE(CodeMat, LibelléMat, CoeffMat)
- EVALUER(#NEtudiant, #CodeMat, Date, Note)

Les données :

1. ETUDIANT

```
mysql> select * from etudiant;
+-----------+--------+---------+
| NEtudiant | Nom    | Prenom  |
+-----------+--------+---------+
|         1 | Rakoto | Aina    |
|         2 | Rabe   | Tiana   |
|         3 | Andry  | Lova    |
|         4 | Rasoa  | Faly    |
|         5 | Jean   | Paul    |
|         6 | Marie  | Claire  |
|         7 | Lucas  | Tom     |
|         8 | Nirina | Mickael |
|         9 | Sarah  | Emma    |
|        10 | David  | Noah    |
+-----------+--------+---------+
10 rows in set (0.00 sec)
```

2. MATIERE

```
mysql> select * from matiere;
+---------+--------------+----------+
| CodeMat | LibelleMat   | CoeffMat |
+---------+--------------+----------+
| MAT1    | Maths        |        4 |
| MAT2    | Physique     |        3 |
| MAT3    | Informatique |        5 |
| MAT4    | Anglais      |        2 |
| MAT5    | Français     |        2 |
+---------+--------------+----------+
5 rows in set (0.01 sec)
```

3. EVALUER

```
mysql> select * from evaluer;
+-----------+---------+------------+------+
| NEtudiant | CodeMat | DateEval   | Note |
+-----------+---------+------------+------+
|         1 | MAT1    | 2026-01-10 |   15 |
|         1 | MAT2    | 2026-01-12 |   14 |
|         1 | MAT3    | 2026-01-15 |   16 |
|         2 | MAT1    | 2026-01-10 |   10 |
|         2 | MAT2    | 2026-01-12 |   12 |
|         2 | MAT3    | 2026-01-15 |   11 |
|         3 | MAT1    | 2026-01-10 |   18 |
|         3 | MAT2    | 2026-01-12 |   17 |
|         3 | MAT3    | 2026-01-15 |   19 |
|         4 | MAT1    | 2026-01-10 |    9 |
|         4 | MAT2    | 2026-01-12 |    8 |
|         4 | MAT3    | 2026-01-15 |   10 |
|         5 | MAT1    | 2026-01-10 |   13 |
|         5 | MAT2    | 2026-01-12 |   15 |
|         5 | MAT3    | 2026-01-15 |   14 |
|         6 | MAT1    | 2026-01-10 |   11 |
|         6 | MAT2    | 2026-01-12 |   10 |
|         6 | MAT3    | 2026-01-15 |   12 |
|         7 | MAT1    | 2026-01-10 |   16 |
|         7 | MAT2    | 2026-01-12 |   15 |
|         7 | MAT3    | 2026-01-15 |   17 |
|         8 | MAT1    | 2026-01-10 |    7 |
|         8 | MAT2    | 2026-01-12 |    9 |
|         8 | MAT3    | 2026-01-15 |    8 |
|         9 | MAT1    | 2026-01-10 |   14 |
|         9 | MAT2    | 2026-01-12 |   13 |
|         9 | MAT3    | 2026-01-15 |   15 |
|        10 | MAT1    | 2026-01-10 |   12 |
|        10 | MAT2    | 2026-01-12 |   11 |
|        10 | MAT3    | 2026-01-15 |   13 |
+-----------+---------+------------+------+
30 rows in set (0.00 sec)
```

• Quel est le nombre total des etudiants ?

```
mysql> SELECT count(*) from ETUDIANT;
+----------+
| count(*) |
+----------+
|       10 |
+----------+
1 row in set (0.00 sec)
```

• Quels sont parmi l'ensemble des notes, la note la plus basse et la note la plus haute ?

```
mysql> SELECT min(Note) AS 'plus basse note', max(Note) AS 'plus haute note' from EVALUER;
+----------------+----------------+
| plus basse note | plus haute note |
+----------------+----------------+
|              7 |             19 |
+----------------+----------------+
1 row in set (0.00 sec)
```

• Quelles sont les moyennes de chaque etudiant dans chacune des matieres ?

```
mysql> SELECT E.NEtudiant, M.LibelleMat, AVG(EV.Note) AS MoyEtuMat
    -> FROM EVALUER EV, MATIERE M, ETUDIANT E WHERE EV.CodeMat = M.CodeMat AND EV.NEtudiant = E.NEtudiant
    -> GROUP BY E.NEtudiant, M.LibelleMat;
+-----------+--------------+-----------+
| NEtudiant | LibelleMat   | MoyEtuMat |
+-----------+--------------+-----------+
|         1 | Informatique |        16 |
|         1 | Maths        |        15 |
|         1 | Physique     |        14 |
|         2 | Informatique |        11 |
|         2 | Maths        |        10 |
|         2 | Physique     |        12 |
|         3 | Informatique |        19 |
|         3 | Maths        |        18 |
|         3 | Physique     |        17 |
|         4 | Informatique |        10 |
|         4 | Maths        |         9 |
|         4 | Physique     |         8 |
|         5 | Informatique |        14 |
|         5 | Maths        |        13 |
|         5 | Physique     |        15 |
|         6 | Informatique |        12 |
|         6 | Maths        |        11 |
|         6 | Physique     |        10 |
|         7 | Informatique |        17 |
|         7 | Maths        |        16 |
|         7 | Physique     |        15 |
|         8 | Informatique |         8 |
|         8 | Maths        |         7 |
|         8 | Physique     |         9 |
|         9 | Informatique |        15 |
|         9 | Maths        |        14 |
|         9 | Physique     |        13 |
|        10 | Informatique |        13 |
|        10 | Maths        |        12 |
|        10 | Physique     |        11 |
+-----------+--------------+-----------+
30 rows in set (0.00 sec)
```

• Quelles sont les moyennes par matieres ? Avec la vue MGETU de la question 3

```
mysql> CREATE VIEW MOYETUMAT AS
    -> SELECT E.NEtudiant, M.LibelleMat, AVG(EV.Note) AS MoyEtuMat
    -> FROM EVALUER EV, MATIERE M, ETUDIANT E
    -> WHERE EV.CodeMat = M.CodeMat
    -> AND EV.NEtudiant = E.NEtudiant
    -> GROUP BY E.NEtudiant, M.LibelleMat;
Query OK, 0 rows affected (0.01 sec)

mysql> show tables;
+------------------------------+
| Tables_in_gestion_des_notes2 |
+------------------------------+
| etudiant                     |
| evaluer                      |
| matiere                      |
| moyetumat                    |
+------------------------------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT LibelleMat, AVG(MoyEtuMat) AS MoyenneMatiere
    -> FROM MOYETUMAT
    -> GROUP BY LibelleMat;
+--------------+----------------+
| LibelleMat   | MoyenneMatiere |
+--------------+----------------+
| Informatique |           13.5 |
| Maths        |           12.5 |
| Physique     |           12.4 |
+--------------+----------------+
3 rows in set (0.00 sec)
```

• Quelle est la moyenne general de chaque etudiant ?

```
mysql> CREATE VIEW MGETU AS
    -> SELECT NEtudiant,
    ->        SUM(MoyEtuMat * CoeffMat) / SUM(CoeffMat) AS MgEtu
    -> FROM MOYETUMAT
    -> GROUP BY NEtudiant;
Query OK, 0 rows affected (0.01 sec)

mysql> show view;
ERROR 1064 (42000): You have an error in your SQL syntax; check the
mysql> select * from MGETU;
+-----------+--------------------+
| NEtudiant | MgEtu              |
+-----------+--------------------+
|         1 | 15.16666666666666 |
|         2 | 10.91666666666666 |
|         3 | 18.16666666666668 |
|         4 |  9.16666666666666 |
|         5 | 13.91666666666666 |
|         6 | 11.16666666666666 |
|         7 | 16.16666666666668 |
|         8 |  7.91666666666667 |
|         9 | 14.16666666666666 |
|        10 | 12.16666666666666 |
+-----------+--------------------+
10 rows in set (0.00 sec)
```

• Quelle est la moyenne générale de la promotion ?

```
mysql> SELECT AVG(MgEtu) AS MoyPromotion
    -> FROM MGETU;
+-------------------+
| MoyPromotion      |
+-------------------+
| 12.89166666666667 |
+-------------------+
1 row in set (0.00 sec)
```

• Quels sont les étudiants qui ont une moyenne générale supérieure ou égale à la moyenne générale de la promotion?

```
mysql> SELECT E.NEtudiant,
    ->        E.Nom,
    ->        E.Prenom,
    ->        M.MgEtu
    -> FROM MGETU M, ETUDIANT E
    -> WHERE M.NEtudiant = E.NEtudiant
    -> AND M.MgEtu >= (
    ->     SELECT AVG(MgEtu)
    ->     FROM MGETU
    -> );
+-----------+--------+--------+--------------------+
| NEtudiant | Nom    | Prenom | MgEtu              |
+-----------+--------+--------+--------------------+
|         1 | Rakoto | Aina   | 15.166666666666666 |
|         3 | Andry  | Lova   | 18.166666666666668 |
|         5 | Jean   | Paul   | 13.916666666666666 |
|         7 | Lucas  | Tom    | 16.166666666666668 |
|         9 | Sarah  | Emma   | 14.166666666666666 |
+-----------+--------+--------+--------------------+
5 rows in set (0.01 sec)
```